

SEMESTER 2 EXAMINATIONS 2023/2024

MODULE: CA208 - Logic

PROGRAMME(S):

COMSCI	BSc in Computer Science
ECSAO	Study Abroad (Engineering & Computing).
ECSA	Study Abroad (Engineering & Computing).
CASE	BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY: 2,O,X

EXAMINERS:	Dr. David Sinclair	(Internal)	(Ext:5510)
	Prof. Arend Rensink	(External)	External

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1

[Total marks: 30]

Consider the following problem.

Anyone passing their logic exam and winning the lottery is happy. But anyone who studies or is lucky can pass all their exams. John did not study but he is lucky. Anyone who is lucky wins the lottery.

Hint: The following predicates may be useful:

- $\text{pass}(x, y)$ is true if x passes subject y .
- $\text{win}(x, y)$ is true if x wins y .
- $\text{happy}(x)$ is true if x is happy.
- $\text{study}(x)$ is true if x studies.
- $\text{lucky}(x)$ is true if x is lucky.

1(a)

[10 Marks]

Convert the statements above into a set of predicate formulae.

1(b)

[10 Marks]

Convert the predicate formulae from Question 1(a) into *Prenex Conjunctive Normal Form (PCNF)*.

1(c)

[10 Marks]

Using General Resolution, prove John is happy.

[End Question 1]

QUESTION 2

[Total marks: 10]

[10 Marks]

Without using truth tables, show that

$$\neg((p \wedge \neg q) \vee (\neg p \wedge \neg q)) \equiv (p \rightarrow q) \wedge (p \vee q)$$

Clearly indicate which equivalence is used at each step.

[End Question 2]

QUESTION 3**[Total marks: 10]**

[10 Marks]

Using *Semantic Tableaux* show the following propositional formula is valid.

$$\vdash ((p \rightarrow q) \rightarrow p) \rightarrow p$$

Clearly indicate which rule was applied at each step.

[End Question 3]**QUESTION 4****[Total marks: 10]**

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \vee (B \wedge C)) \rightarrow ((A \vee B) \wedge (A \vee C))$$

Clearly indicate which rule was applied at each step.

[End Question 4]**QUESTION 5****[Total marks: 10]**

[10 Marks]

Using Gentzen system $\mathcal{G}$, prove the following predicate formula.

$$\forall x.(p(x) \rightarrow q(x)) \rightarrow (\forall x.p(x) \rightarrow \forall x.q(x))$$

Clearly indicate which rule was applied at each step.

[End Question 5]**QUESTION 6****[Total marks: 10]**

[10 Marks]

Using *Skolemisation*, convert the following predicate expression into a set of clauses.

$$\forall x.((\exists y.R(x, y) \wedge \forall y.\neg S(x, y)) \rightarrow \neg(\exists y.R(x, y) \wedge P))$$

[End Question 6]

Section B

QUESTION 7

[Total marks: 10]

[10 Marks]

Write a Prolog predicate `oddsum(X, S)` that is true when `S` is the sum of all the odd integers in the list `X`. Note that the list `X` can contain any atom, e.g. `[4, oranges, 5, 2, 7, apples, 1]`. Hence, `oddsum([4, oranges, 5, 2, 7, apples, 1], 13)` is true.

[End Question 7]

QUESTION 8

[Total marks: 10]

[10 Marks]

Write a Prolog relation `split(N, Positive, Negative)` that is true if all the positive numbers (including 0) in the list `N` are in the list `Positive`, and all the negative numbers in the list `N` are in the list `Negative`.

[End Question 8]

QUESTION 9

[Total marks: 10]

[10 Marks]

Describe the procedural interpretation of a Prolog program. Provide a sample Prolog query and show how this procedural interpretation evaluates the query.

[End Question 9]

APPENDICES

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]